Indian scholars of the 5th century AD(CE), realized that working with half-chords for half-angles greatly simplified the theory of chords and its application to astronomy. Mathematicians like Aryabhata, the two Bhaskaras and several others developed astonishingly sophisticated techniques for calculating half-chord (Jya) values.

Mathematician Abu Al-Wafa of Baghdad believed to have invented the tangent function, which he called the "Shadow". Arabic scholars did not know how to translate the word Jya, into their texts and simply wrote jiba as a close approximate word.

Misinterpreting the Arabic word 'jiba' for 'cove' or 'bay', translators wrote the Arabic word 'jiba' as 'sinus' in Latin to represent the half-chord. From this, we have the name 'sine' used to this day. The word "Trigonometry" itself was invented by German mathematician Bartholomaeus Pitiscus in the beginning of 17th century AD(CE).

Recall

Trigonometric Ratios

Let $0° < \theta < 90°$

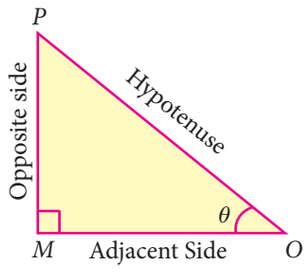


Fig. 6.1

Let us take right triangle OMP

$$\sin\theta = \frac{\text{Opposite side}}{\text{Hypotenuse}} = \frac{MP}{OP}$$

$$\cos\theta = \frac{\text{Adjacent side}}{\text{Hypotenuse}} = \frac{OM}{OP}$$

From the above two ratios we can obtain other four trigonometric ratios as follows.

$$\tan\theta = \frac{\sin\theta}{\cos\theta};\ \cot\theta = \frac{\cos\theta}{\sin\theta};$$
$$\text{cosec}\theta = \frac{1}{\sin\theta};\ \sec\theta = \frac{1}{\cos\theta}$$

Note

All right triangles with θ as one of the angle are similar. Hence the trigonometric ratios defined through such right angle triangles do not depend on the triangle chosen.

Trigonometric ratios of complementary angle

$\sin(90° - \theta) = \cos\theta$	$\cos(90° - \theta) = \sin\theta$	$\tan(90° - \theta) = \cot\theta$
$\text{cosec}(90° - \theta) = \sec\theta$	$\sec(90° - \theta) = \text{cosec}\,\theta$	$\cot(90° - \theta) = \tan\theta$

Visual proof of trigonometric complementary angle

Consider a semicircle of radius 1 as shown in the figure.

Let $\angle QOP = \theta$.

Then $\angle QOR = 90° - \theta$, so that $OPQR$ forms a rectangle.

From triangle OPQ, $\frac{OP}{OQ} = \cos\theta$

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